

Week 7 Tutorial

Idea Generation

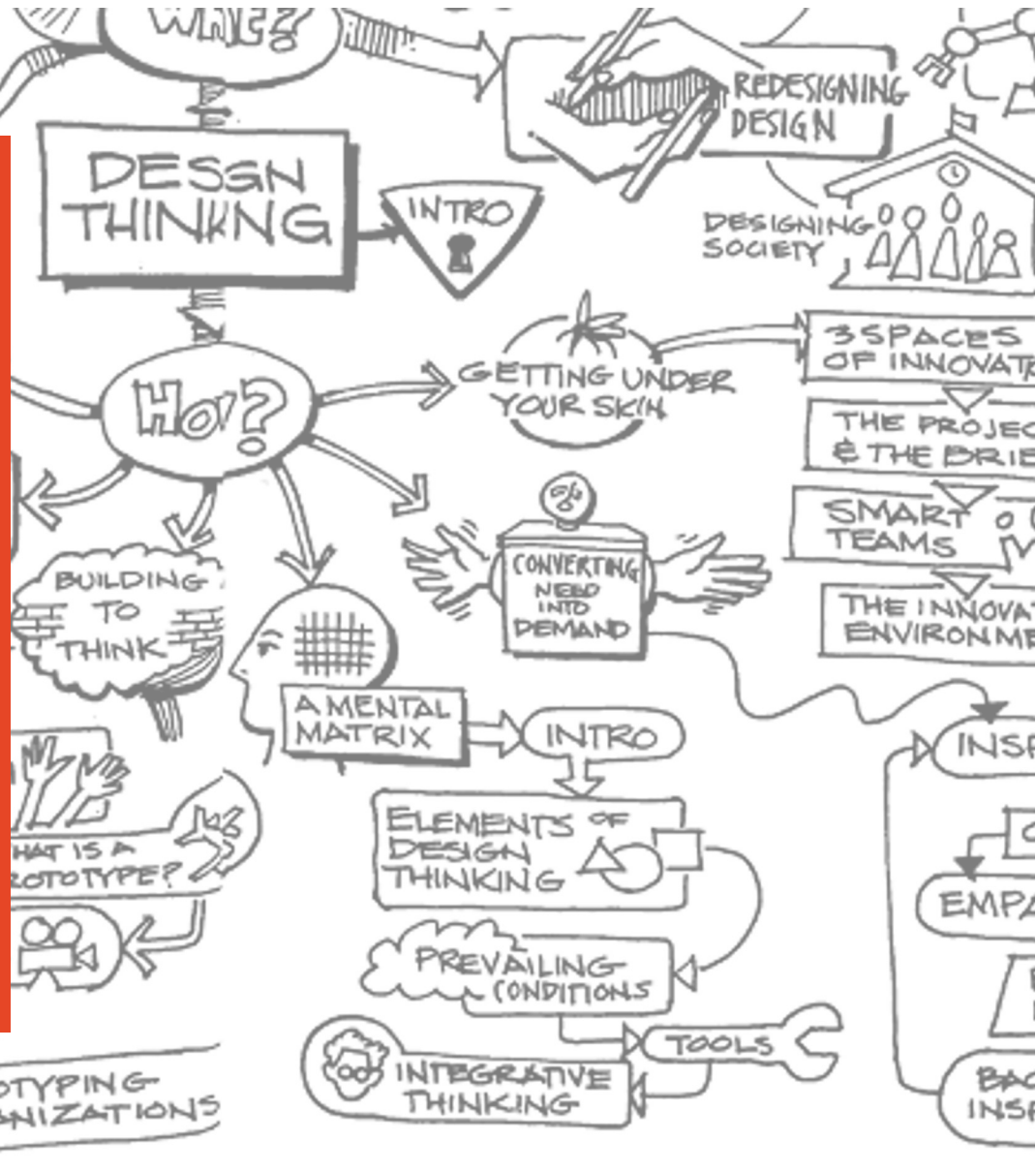
Lo-fi Prototypes

IDEA9106 Design Thinking
Semester 2 2026



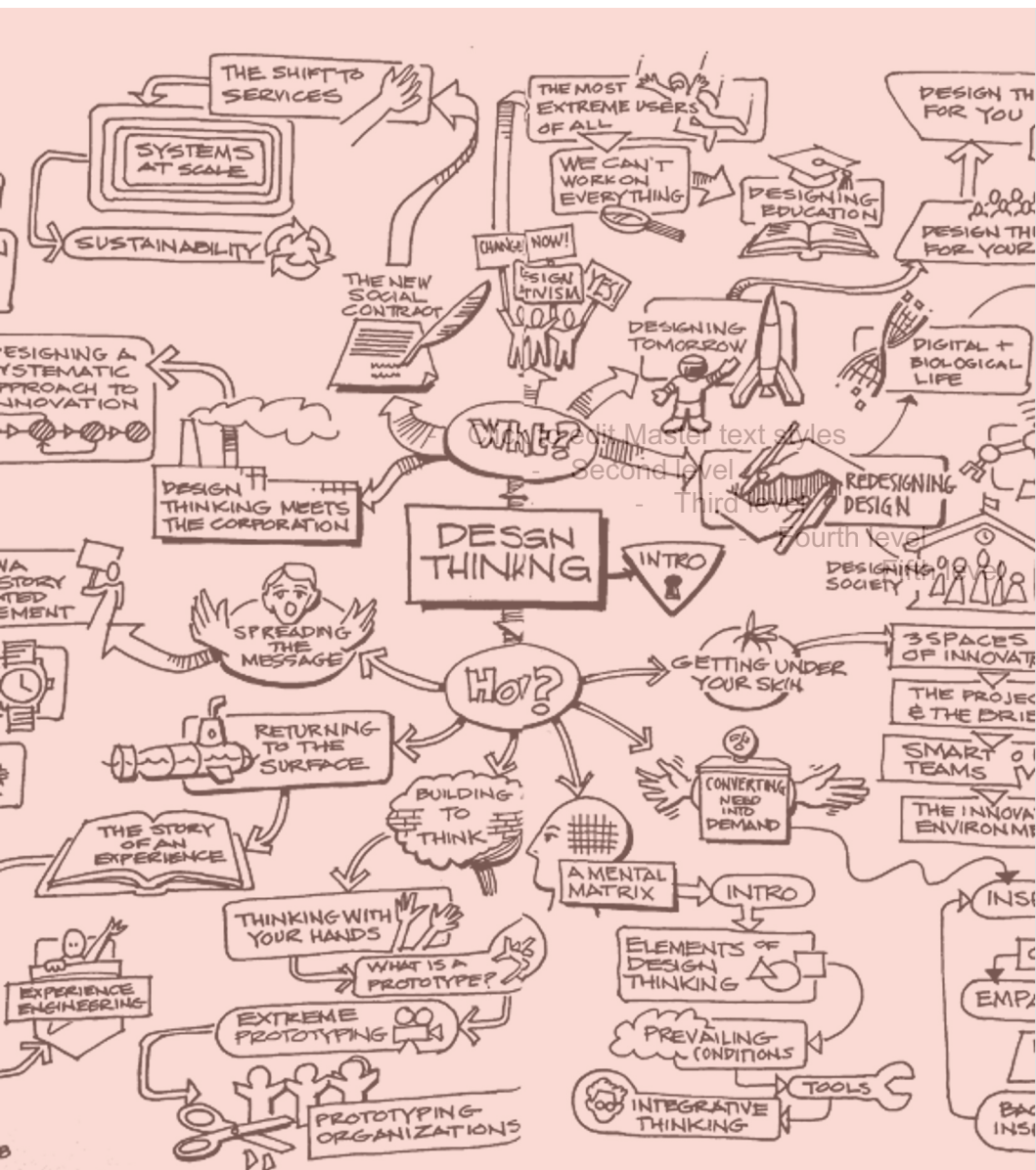
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We recognise and pay respect to the Elders and communities - past and present - of the lands upon which this University of Sydney campus sits. Indigenous knowledge has been generated and shared for thousands of years, underpinned by a strong connection to land, sea and culture. We seek to continue to exchange knowledge and build connections upon these traditions.



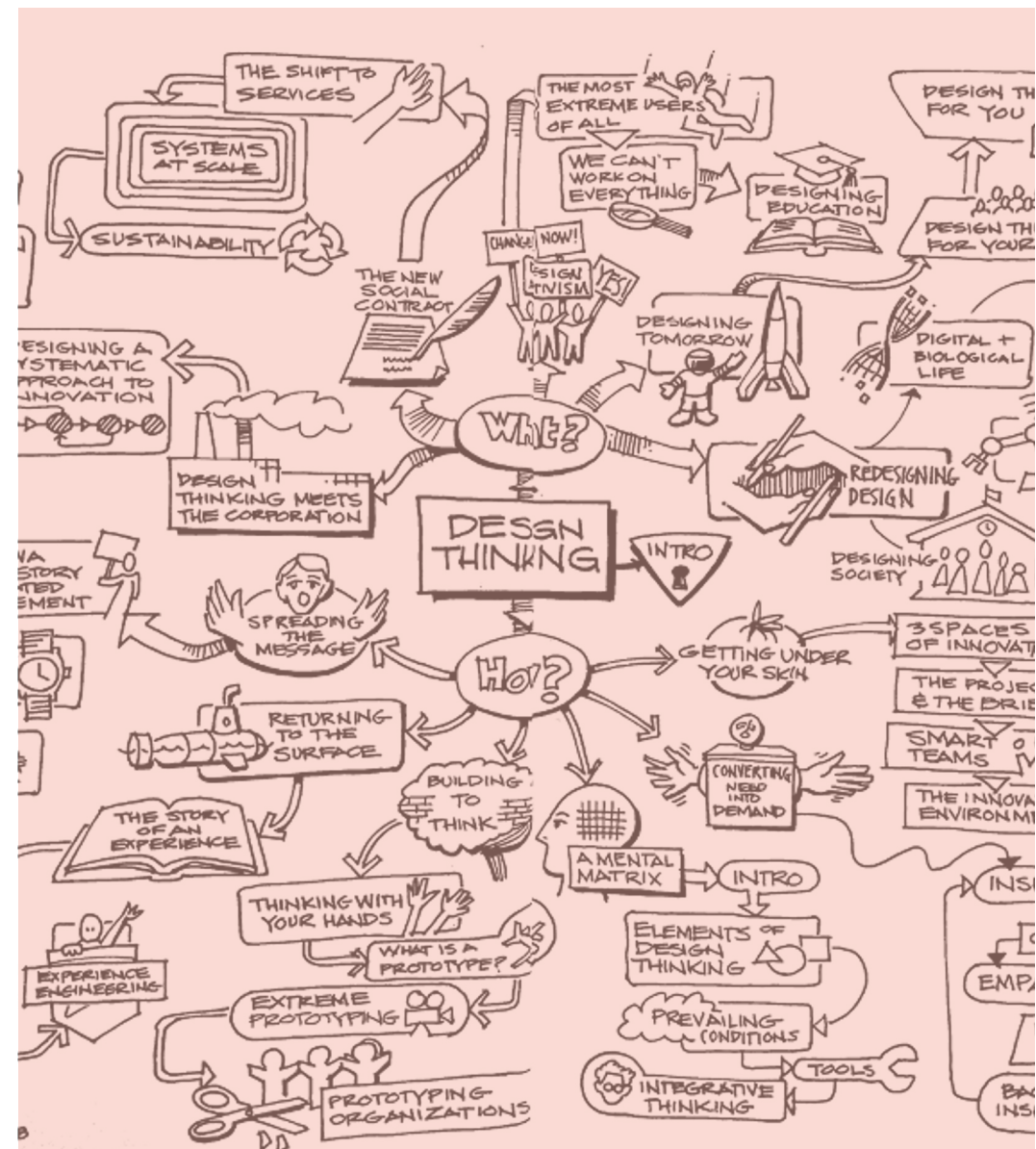


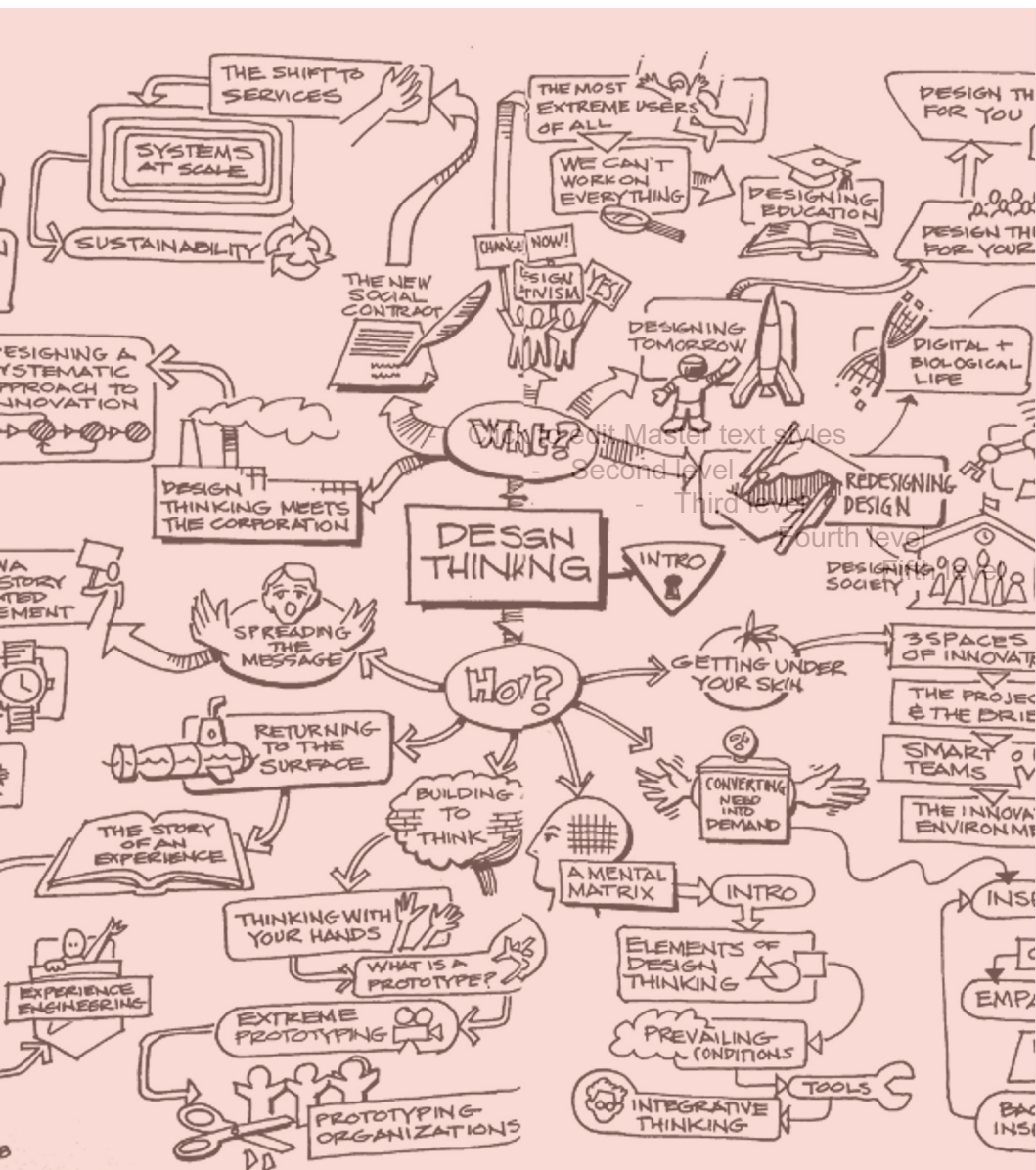
Tutorial Aim

In this tutorial students will:

- generate ideas using KJ Brainstorming, Brainwriting 6-3-5, Bodystorming, Mind Mapping and/or the Worst Possible Idea method
- discuss, iterate and choose the 'best' idea
- plan/begin lo-fi prototyping

- **Generate ideas**
- **Lo-fi prototyping**
- **Wrap Up**





Generate ideas

In this activity groups will generate evidence-based ideas informed by their project group's user data, personas, reframed design brief, problem/opportunity space



Generate ideas (50 mins)

- Read through the idea generation methods on Figma
- Choose the most appropriate method for addressing the group's reframed project brief. Hot tip: Bodystorming will be relevant for many of the groups.
- Complete the most appropriate method

If you have arrived today with an idea that you believe to be an evidence-based solution for the group's reframed project brief, please let me know so I can give feedback



Part 1: Generating Ideas

- Generate as many ideas as possible, using several different methods, quantity over quality at this stage
 - As a group, do a quick discussion-based evaluation: cluster similar ideas, flag anything genuinely promising
 - Use Worst Possible Idea deliberately: sometimes the fastest way to a good idea is naming the bad one first
 - Once you have a few promising directions, move into prototyping to test them
-
- **Five methods today:** Brainstorming, Brainwriting 6-3-5, Bodystorming, Mindmapping, Worst Possible Idea



Brainstorming

A group generates as many ideas as possible out loud, in a short, high-energy burst, with **judgement** and **evaluation** deliberately **switched off** until the burst ends.

It works because generating ideas and judging them use different mental modes. Judging an idea while it's still forming shuts down the thought that would have come next, so separating the two phases lets quantity build its own momentum: one idea triggers another, and more raw material increases the odds that something genuinely new turns up among the obvious answers.

The rule that makes it work: no criticism, no feasibility checks, no 'yes, but', only 'yes, and'. Every idea gets recorded, even the silly ones, since a silly idea can still carry a seed worth extracting later.



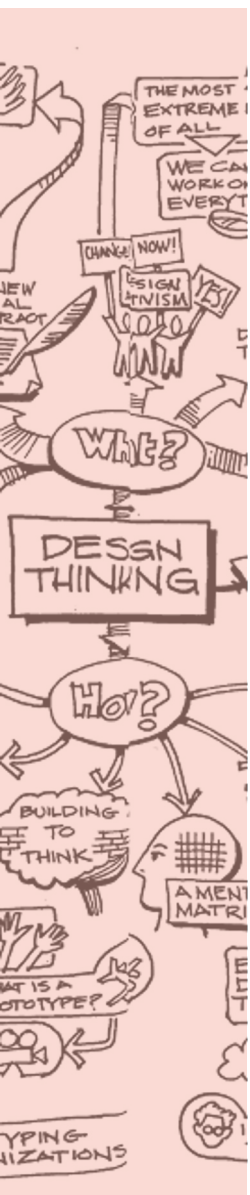
Brainstorming

Bad example

- One person suggests an idea, the group immediately debates whether it's feasible before the next idea can surface
- Four ideas come out in ten minutes, because each one gets picked apart on the spot

Good example

- Every idea goes straight onto its own sticky note, no discussion and no feasibility check until time is up
- Forty raw ideas come out in ten minutes, because nobody was allowed to say 'that won't work' yet

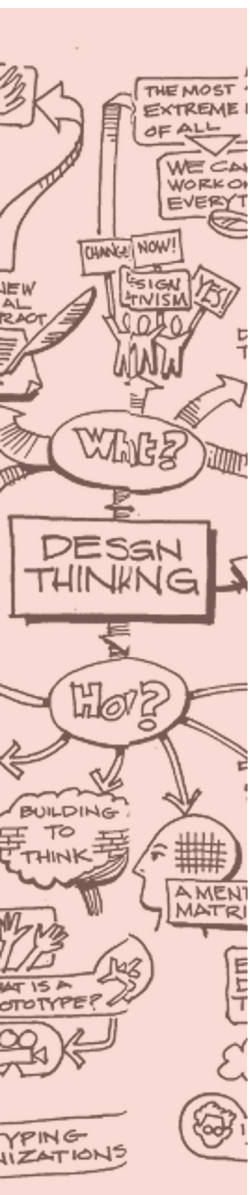


Brainwriting

A structured, mostly-silent alternative to spoken brainstorming. In its classic form, six people each write three ideas in five minutes, then pass their sheet on. The next person reads what's already there and adds three more ideas that build on it, and the sheets keep circulating until everyone has contributed to every sheet.

It works because ordinary group brainstorming favours whoever speaks first, loudest, or holds the most status in the room. Writing in silence removes that dynamic: every participant produces the same volume of ideas regardless of personality, and reading someone else's idea before adding your own means new ideas build on what exists instead of everyone starting from a blank page.

Because it runs in parallel rather than one idea at a time, brainwriting can generate far more ideas in the same amount of time than spoken brainstorming.



Brainwriting

Bad example

- The team skips the silent-writing step and just goes around the table saying ideas out loud in turn
- It becomes ordinary brainstorming again, and the loudest person's first idea anchors everyone else's variation on it

Good example

- Each person privately writes three ideas in silence, then passes the sheet on without saying whose ideas were whose
- A quieter teammate ends up contributing as many ideas as the most talkative person in the room



Bodystorming

A physical, embodied form of brainstorming: acting out a scenario with your body and the space around you instead of describing it seated at a table.

Often uses simple props, improvised roles (an actor living the scenario, an observer documenting it), and rough physical mock-ups of the space or objects involved.

It works because much of what shapes an experience, posture, timing, what's within reach, what's visible from where you're standing, is tacit. People don't consciously mention these things when describing a situation out loud, because they aren't aware they're relying on them. Physically inhabiting the scenario surfaces this tacit knowledge: your body notices constraints and frictions that a verbal description of the same scenario would have skipped straight past.

Especially useful when physical, spatial or timing details matter to the problem, and works well as a warm-up before moving into higher-fidelity prototyping.



Bodystorming

Bad example

- The team stays seated and describes what a person in a queue would probably feel
- It's a thought experiment, not a bodystorm, so no new physical insight comes out of it

Good example

- Two students physically stand in a slow line while carrying groceries and checking a phone
- They discover the real friction isn't the wait time, it's not knowing when to look up, an insight talking never would have surfaced



Mindmapping

A visual note-making technique that radiates outward from a central topic using branches and keywords, often structured around the prompts who, what, where, when, why and how, to build a fast, wide overview of a problem area.

It works because linear written notes hide relationships between ideas, while a mindmap uses position and connection to show how the pieces of a problem relate to each other.

Gaps become visible at a glance: an empty branch is a question nobody has answered yet.

Structuring branches around a fixed set of prompts (eg what, when, where, why, how, who) also forces the same problem to be looked at from several angles, rather than only the one the team defaults to.



Mindmapping

Bad example

- The map is really a to-do list with lines drawn between boxes, e.g. app → needs login → needs payment
- No branch reveals anything new, it just repeats what the team already knew going in

Good example

- The team puts 'food delivery for share-house students' in the centre and works through all six branches properly
- The 'when' branch reveals most orders happen between 9pm and midnight, when nothing on campus is open, a constraint nobody had named yet

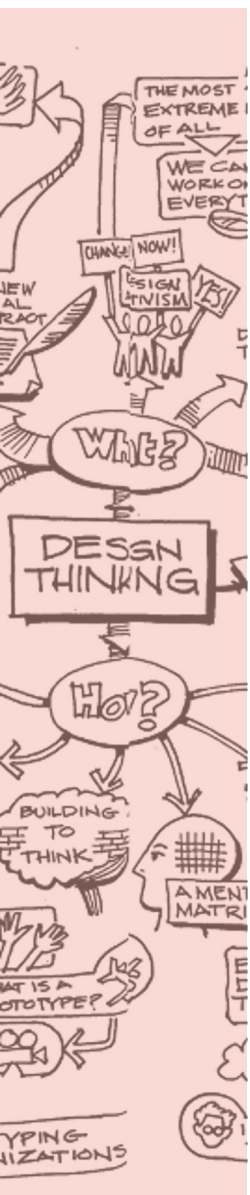


Worst Possible Idea

Deliberately generating the worst, most terrible and impractical ideas the team can think of, on purpose, then examining what each one reveals and inverting it to find a genuinely workable direction.

It works because groups often get stuck when people are quietly self-censoring, holding an idea back because it might sound naive, obvious or embarrassing. Giving explicit permission to be bad removes that fear entirely, nobody has to defend a worst idea, it's supposed to be bad, and this usually unlocks a looser, more playful mode of thinking that produces genuinely novel directions once inverted.

The distinction that matters: a good worst idea is revealing, not just impractical, it should expose an assumption the team didn't know it was making, and from that what to avoid in a good idea. Once a batch of worst ideas exists, invert each one and ask what real principle or need it points to.



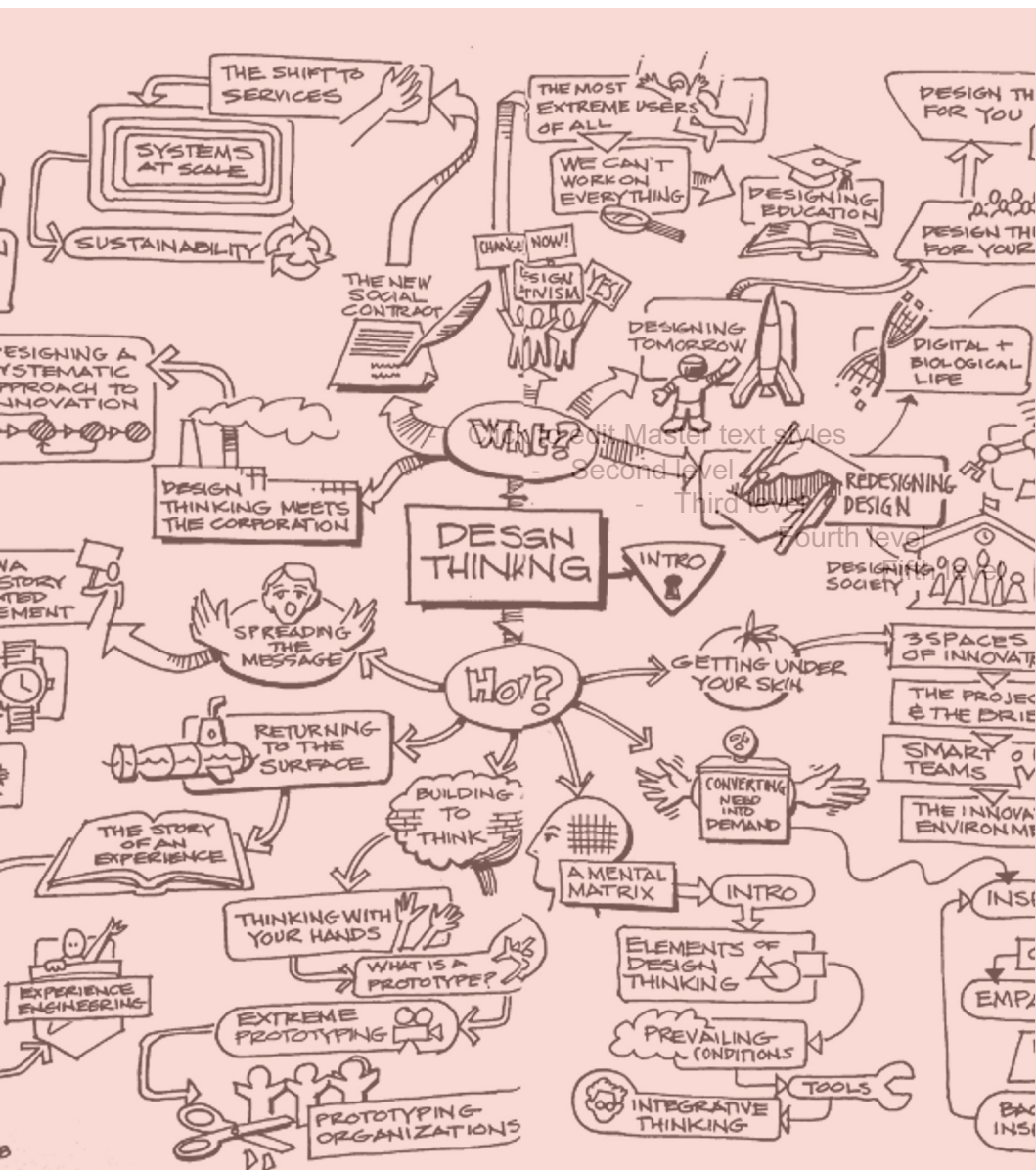
Worst Possible Idea

Bad example

- The team writes ideas that are just impractical rather than revealing, e.g. 'make the app really slow'
- Inverting it just gives 'make the app fast', which was already obvious before the exercise

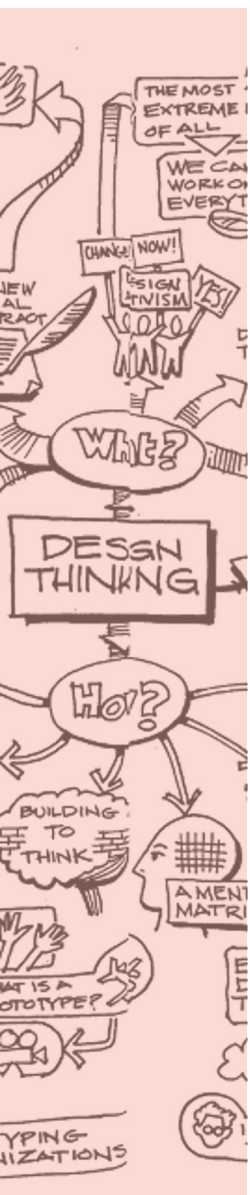
Good example

- For an app to help elderly users manage medication, someone suggests 'an app that only works in a language the user doesn't speak'
- Inverting it reveals a real principle: default to whatever language the phone is already set to, no separate onboarding step



Lo-fi prototyping

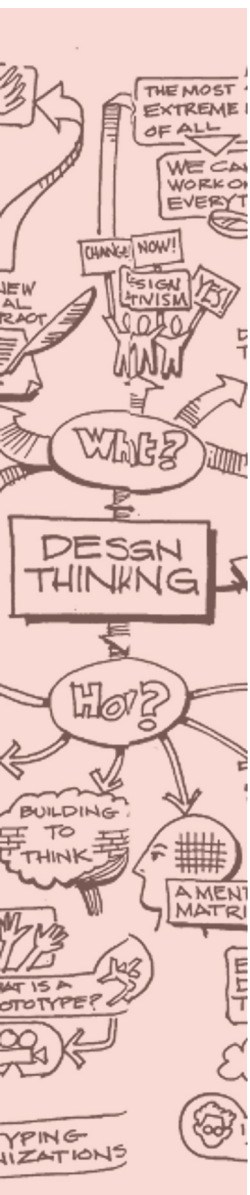
In this activity groups will progress their 'best' idea and plan/create a lo-fi prototype



Lo-fi prototyping (50 mins)

As you work, please keep in mind the requirements of A2d. Final Presentation, that indicates prototypes must:

- respond to the group's reframed brief and the research that has been undertaken to date
- be effective for testing and enable the group to validly assess a users' experience and/or stakeholder assessment of the design concept
- have the capacity to be modified/iterated in response to user testing feedback



Evaluate the idea:

Have you evaluated all ideas as a project group and iterated the 'best' idea?

If the answer is '**No**':

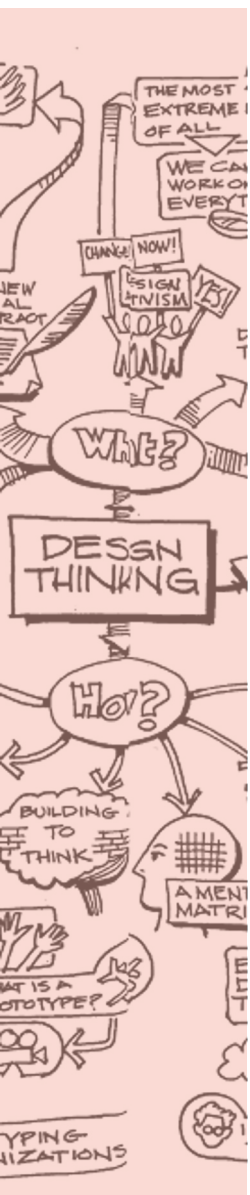
- spend 15 mins iterating your idea

If the answer is '**Yes**':

- you are ready to start prototyping using the resources on Figma

Don't forget your stakeholders!!!

An app or technical system that gives clients information on their wait might be expensive, cumbersome and generate extra tasks for workers. Make sure you evaluate the impact on your two primary stakeholders – those waiting and those controlling the wait.



With your 'best' idea as the starting point:

- Sketch the parts of the idea that you want to represent in the prototype (approx. 10mins)
- Create sketches of key interactions (approx. 10mins)
- Create a lo-fi paper prototype based on the sketches, including real content for key interactions (approx. 20mins)
- Answer the three rationale questions on Figma to help reflect on the appropriateness of the prototype (approx. 10mins)



Prototyping

The purpose of prototyping is to make a platform to test whether your solution actually solves the underlying problem.

- **A prototype is not the deliverable**, it's a mechanism for asking a real question. Before you build anything, write down exactly what question you're testing, and what result would mean the idea has failed
- If your idea happens to be an app, the app is just the delivery mechanism, **what you're actually testing is what's underneath it**

The App Trap: many of you will land on 'build an app' as your solution. That's fine, apps are often a reasonable delivery mechanism

- The trap is prototyping the app instead of the idea inside it. Testing button placement tells you nothing about whether the underlying idea works



Prototyping App-Based Ideas: Guidelines

- **Name the mechanism first.** Write one sentence: 'this solves the problem by ____.' If it still makes sense with the word 'app' deleted, you're ready to prototype.
- **Prototype the behaviour, not the navigation.** Fake the 'smart' part, the algorithm, the matching, the recommendation, with a human or a spreadsheet instead of trying to build it (Wizard of Oz).
- **Test the task, not the tour.** Don't ask 'can you find the settings menu', ask 'does this actually get you to take your medication on time.'
- **Low fidelity is a feature.** A rough paper screen forces testers to react to the concept, not the aesthetics. A polished prototype gets critiqued on fonts, not function.



Prototyping App-Based Ideas: Guidelines

AI does not prototype

Some of you may be tempted to prompt Gen-AI to make a 'prototype' for you, but there are many issues with that.

The process of generating an App walkthrough or prototype with AI does not create tests for your idea as a solution, which is the aim of prototyping.



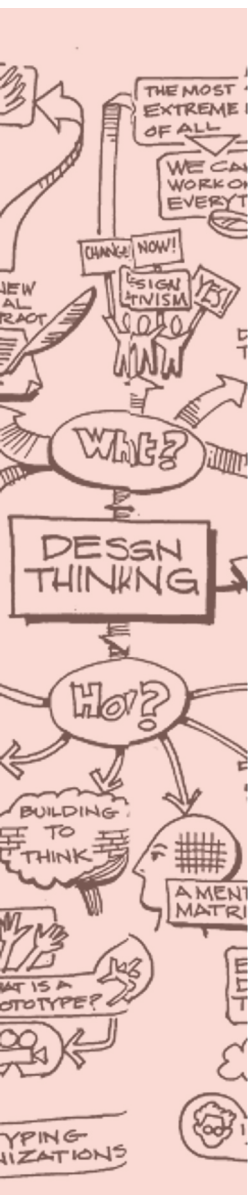
Low-fidelity Prototyping

A rough, quickly-made representation of an idea, built from paper, cardboard or other simple materials, used to explore a concept before investing time in a polished version.

It works because a prototype that so obviously isn't finished invites people to engage with the idea underneath it rather than its finish. It's fast and cheap enough to change that it can be built, tested and rebuilt several times in one session, revealing whether the core concept holds up long before it would be worth polishing anything.

What it tests: whether the underlying idea, the process, the logic, the value it offers someone, actually works, not what the final product will look like.

A real test asks someone to attempt a genuine task with it, and watches for where the idea itself breaks down, not where the paper is awkward to hold.



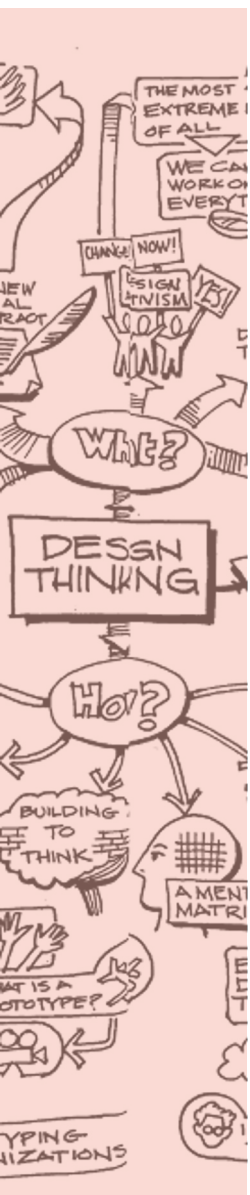
Low-fidelity Prototyping

Bad example

- The team spends the whole session drawing a polished-looking home screen in ink, complete with a logo
- The 'test' becomes 'does this look like a real app', which says nothing about whether the underlying idea works

Good example

- A 'split the bill fairly' concept is built from paper slips and a hand-drawn receipt, then tested on friends actually splitting a real dinner bill
- They learn people forget to log shared items like the appetiser, a genuine insight about the process being designed

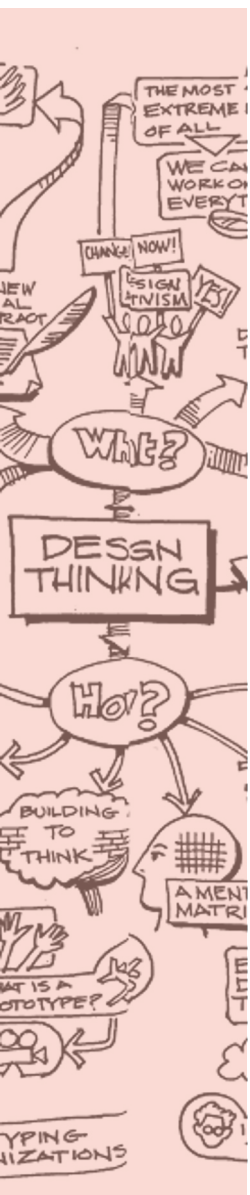


Wireframing

Simple line drawings that show only the structure and functional layout of a system, screens, menus, buttons, and the flow between them, with no colour, imagery, branding or typography included.

It works because stripping away every visual decision forces attention onto structure alone: does the information architecture make sense, can someone get from A to B in a sensible number of steps, is anything missing or duplicated. Because wireframes are quick to draw and redraw, they invite structural changes, it is easy to to cross out a box and redraw it.

What it tests: the logical structure and flow of an idea, not whether the screens look appealing. If a wireframe review turns into a conversation about colours or fonts, it has too much visual detail in it and isn't doing its job.



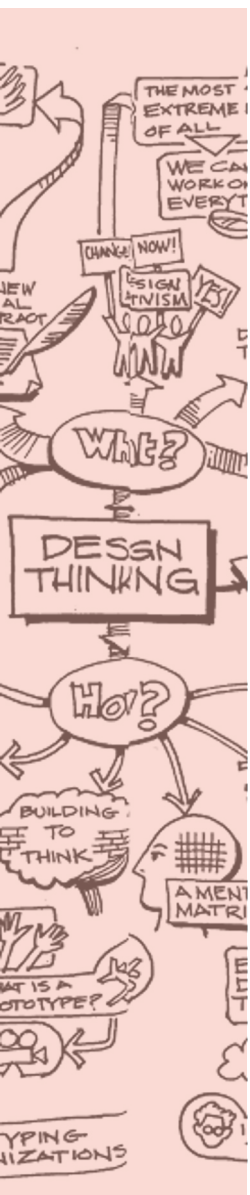
Wireframing

Bad example

- The 'wireframe' includes downloaded app icons, brand colours and a mocked-up loading animation
- The review turns into a conversation about whether the colour scheme feels trustworthy, not whether the structure makes sense

Good example

- Boxes and arrows only, checking whether a user reaches 'next alert' in the fewest taps for a transport-delay idea
- A 15-minute partner review focuses entirely on structure, because there's no visual noise to distract from it



Experience Prototyping

A life-size, physical re-creation of a situation, built from simple materials and space, that people can actually move through, rather than look at on a screen or read about on paper.

It works because some of what an idea needs to get right, how long a wait actually feels, whether an interaction feels reassuring or stressful, whether people can find each other in a space, can only be judged by being in the situation, not by describing it. Building the physical situation and having real people move through real scenarios in it surfaces exactly these experiential and temporal qualities.

What it tests: the lived experience around an idea, timing, emotion, social dynamics, physical comfort, not the interface that might eventually deliver it.



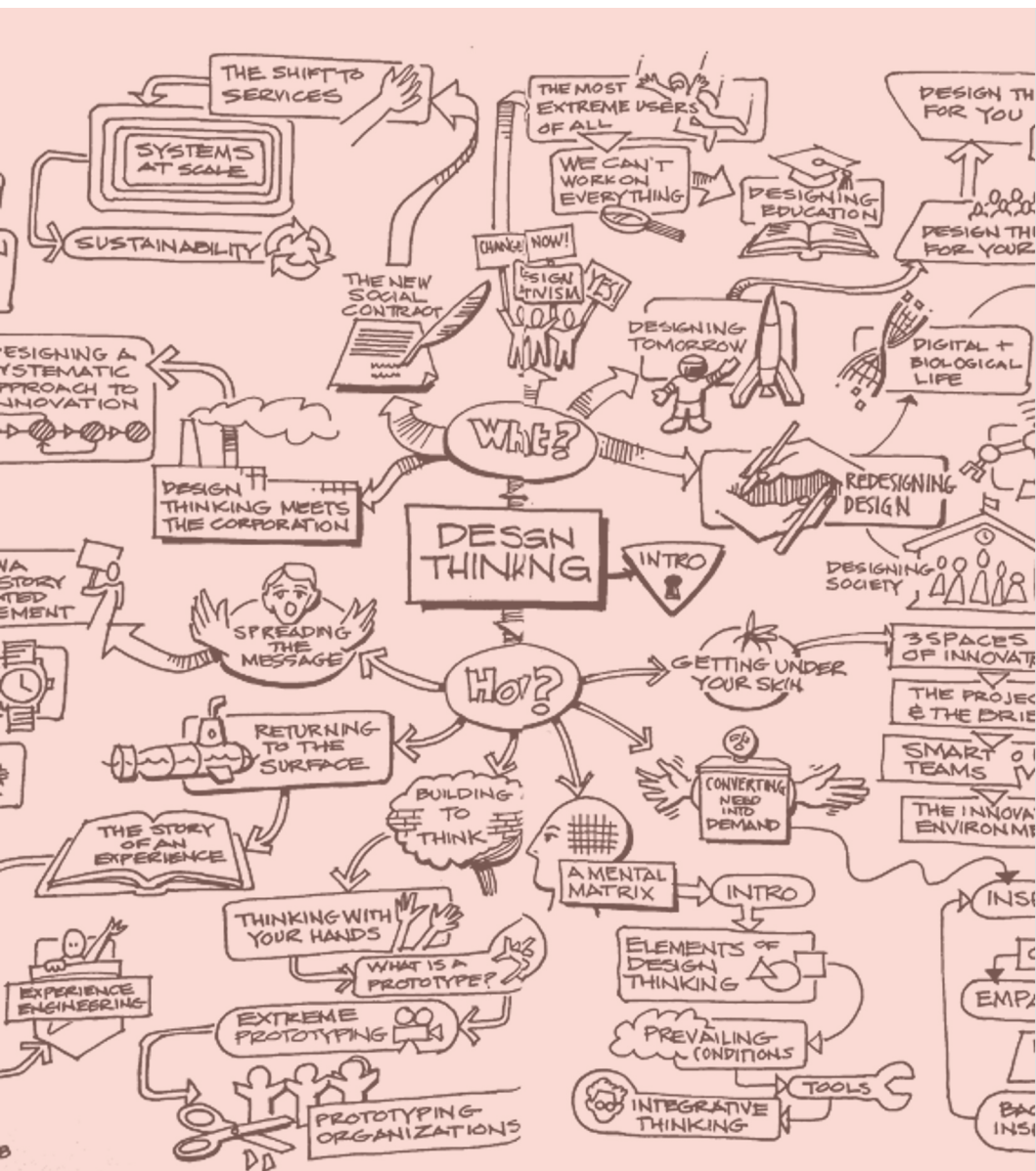
Experience Prototyping

Bad example

- The team 'experience prototypes' a queue by describing the scenario verbally while seated
- Nobody actually waits, so nobody experiences the boredom or the phone-checking the idea is meant to address

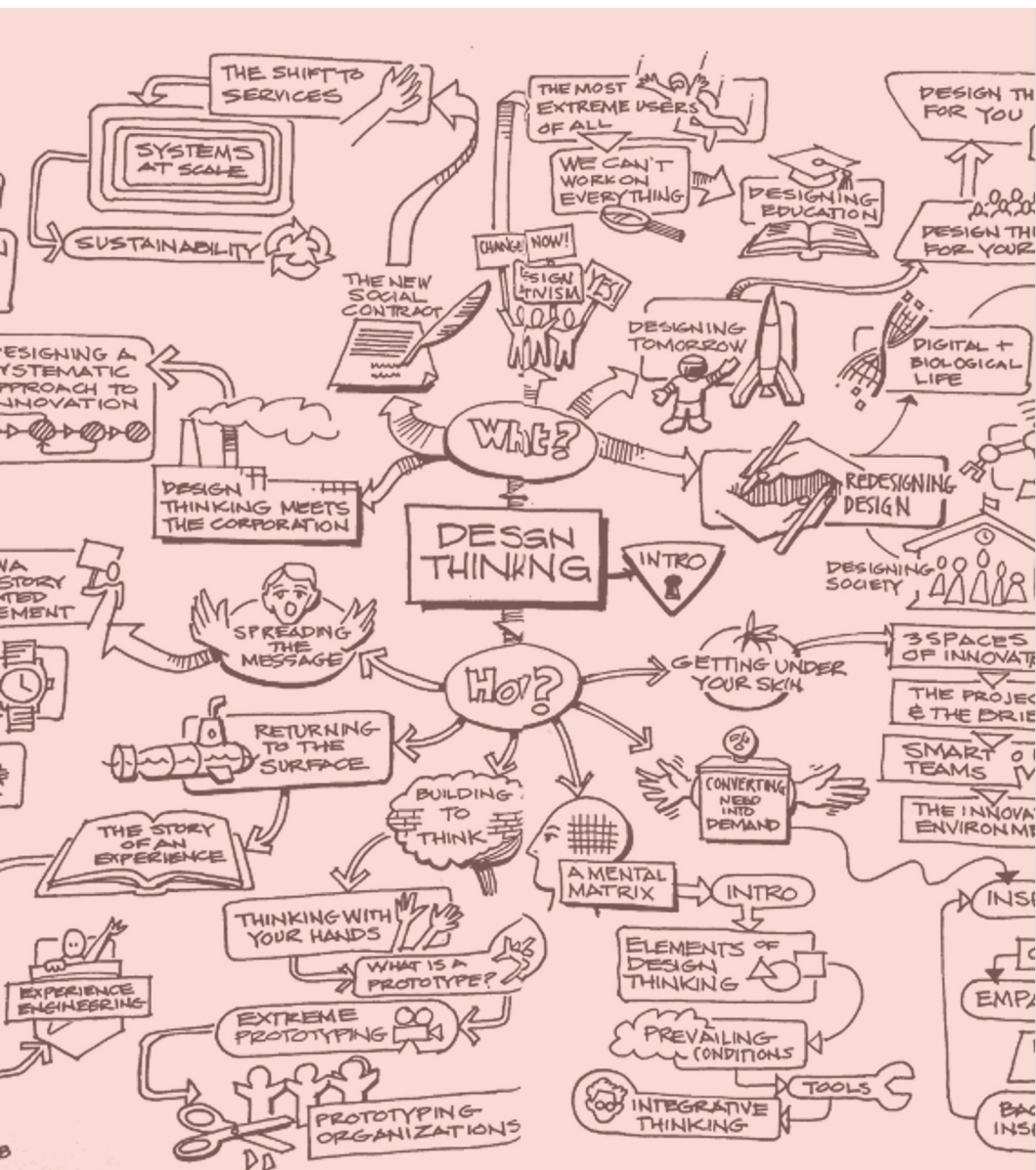
Good example

- For a smart-queue idea, the team tapes out a queue on the floor and hands out a paper ticket instead of a phone screen
- Classmates who actually wait learn that 90 seconds feels different depending on whether you can see progress, an insight about experience, not a screen



Wrap Up

- Any unexpected insights from today's tutorial?
- Anything that's still not clear or is confusing?
- Any questions or thoughts?



Before next week

- Continue to develop and iterate your prototype ready for showcasing at the progress presentations next week during tutorials
- Prepare A2b. Progress Presentations
- Read the Week 8 module & complete the pre-work

SEE YOU NEXT WEEK!



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